

Prikshit Gautam

✉ pgautamlinkedin@gmail.com — 📞 (+91) 9877035742 — 📍 Patiala, Punjab
🐙 GitHub — 🔗 LinkedIn — 🌐 Portfolio — 📝 Medium

Education

Thapar Institute of Engineering and Technology, Patiala

B.E. Electronics and Computer Engineering, CGPA: 8.56

Aug 2023 – Jun 2027

St. Soldier Divine Public School, CBSE, Nangal

XII : 92.67%

May 2022

Technical Skills

Languages: C++, Python, JavaScript, SQL

Libraries & Frameworks: PyTorch, Scikit-learn, TensorFlow, Keras, ONNX Runtime, LangChain, FAISS, Flask, NumPy, Pandas, Matplotlib, PyMuPDF

Developer Tools: AWS (EC2, S3), Render, Vercel, Git, GitHub (CI/CD), Postman, Streamlit, Hugging Face

AI Technology: NLP, RAG, LLM Integration, Vector Search, Time Series Forecasting, Model Optimization, Data Visualization

Databases: MySQL, DynamoDB

Relevant Coursework: Database Management Systems, DSA, Object Oriented Programming, Cloud Computing, OS, Computer Networks, Operating System

Experience

Undergraduate Research Assistant | TIET, Patiala

Aug 2024 – May 2025

Project: 5G IP Latency Predictor using Deep Sequential Methods | [LINK](#) | [LIVE](#)

Technology: PyTorch, Keras, Statistical Analysis

- Analyzed **40,000+** 5G packet traces from the **KTH Expeca testbed (Sweden)**, isolating RLC frame alignment limitations as a key structural source of network delay.
- Designed a hybrid LSTM model forecasting 5G latency with **94.68%** tracking accuracy (R^2 : **0.91**), mitigating multi-step error accumulation; co-authored a journal paper titled **"Forecasting 5G Delay Using Deep Sequential Methods"** under review.
- Reduced multi-step prediction variance by **45%** over baseline autoregressive variants using a custom sequential modeling loop configured for cellular edge environments.
- Optimized deployment via **tf2onnx**, dropping RAM from **450MB to 80MB**; deployed a Flask REST API for a model that predicts batch payloads of **3,470 packets in 2.5 seconds** via an $O(N)$ execution loop.

Experiential Learning Centre | Summer Project Intern | [Credentials](#)

Jun 2025 – Jul 2025

- Led development of an automated animal-hazard visual recognition module using YOLOv8, OpenCV, and SUMO/TraCI to mitigate roadway traffic hazards.
- Coordinated system integration across traffic components, validating system logic matrices before a panel of 10+ faculty reviewers.

Projects

Healthcare Assistant Chatbot (MediQuery) | [GitHub](#) | [LIVE](#)

Jan 2025 – Feb 2025

Technology: Python, LangChain, FAISS, Groq LPU, Streamlit

- Built a healthcare RAG pipeline parsing a **759-page** encyclopedia into **7,470** chunks using **PyMuPDF** and **RecursiveCharacterTextSplitter**.
- Embedded text via **all-MiniLM-L6-v2** with **L2 normalization** for **40ms** CPU vector retrieval across **3.1M** characters.
- Streamed context passages to **Llama-3.1-8B-Instant** via **Groq LPUs** (temp **0.2**), delivering fact-grounded medical explanations in under **1.2 seconds**.
- Selected for an elite enterprise accelerator co-sponsored by **Microsoft & SAP** via Edunet Foundation.

AI-RoadIntelligence and Management System | [LINK](#)

Jun 2025 – Jul 2025

Technology: YOLOv8, OpenCV, SUMO, SightEngine API

- Narrowed a YOLOv8 80-class detector to **2** safety-critical classes (cow, horse) to isolate large road-blocking hazards.
- Constructed a frame-sampling pipeline running inference every **5th frame**, cutting compute overhead by **80%** with automated CSV logging.
- Integrated detections with a **SUMO/TraCI** controller, applying a **20s** presence threshold for alerts and a **60s** green-phase extension.

Achievements & Certifications

- Adobe Hackathon — Top 3.5% Globally (Ranked 4,029/115,000 Teams)** [\[Certificate\]](#)
- Complete Data Science and Machine Learning(under instructor Krish Naik (Udemy))** [\[Certificate\]](#)
- Microsoft Azure AI Fundamentals: Generative AI** [\[Certificate\]](#)
- Data Structures & Algorithms Proficiency** Ongoing
Solved **450+** **DSA** problems to optimize runtime complexity across competitive platforms like [LeetCode](#), [NeetCode](#) and [GeeksForGeeks](#).